

HOÀNG QUỐC TÂN

Full-Stack Software Engineer · Systems Design · AI-Integrated Applications

+84 328-728-617 | hoangquoc.tan.1996@gmail.com | github.com/tanhoang0803 | linkedin.com/in/tanhoang080396 | leetcode.com/u/tanhoang0803

PROFESSIONAL SUMMARY

Fresher full-stack engineer with a senior-architect mindset — I think in systems, not just features. Practical experience spans enterprise banking internships (TD Bank, RBC), a developer role at Infosys building .NET and Node.js platforms, and two independently built production applications integrating AI validation, IoT predictive analytics, and automated CI/CD pipelines.

FLAGSHIP PROJECTS

The Clean Label E-Commerce Engine *React 18 · Redux · Node.js · Express · PostgreSQL · OpenAI GPT-4 · Stripe · Docker · GitHub Actions*

[Live Demo](#) | [GitHub](#)

- Architected AI as a mandatory validation gate — every product triggers a GPT-4 ingredient audit before hitting the database; non-compliant products are blocked automatically.
- Designed Controller → Service → Repository layered architecture with raw SQL (no ORM) — each layer independently testable, query plan fully transparent.
- Built Stripe Checkout + Webhook with HMAC-SHA256 signature verification via `express.raw()` middleware, preventing tampered payment payloads.
- Shipped a 4-job CI/CD pipeline (GitHub Actions → Render + Vercel): lint, Jest/Supertest tests, Docker multi-stage build, auto-deploy on main.
- Persisted AI audit results (`is_safe`, `ai_reason`) at write-time — GET endpoints are pure SQL with zero OpenAI calls, eliminating read-time latency and cost.

Smart-Fleet: Predictive IoT Maintenance Engine *React · Redux Toolkit · Vite · Node.js · Express · Supabase · Upstash Redis · OpenWeather · Firebase FCM · Docker*

[Live Demo](#) | [GitHub](#)

- Designed a multi-factor wear algorithm combining OpenWeather humidity/temperature with mileage telemetry and E10 fuel coefficients — adjusting maintenance intervals by up to 53% under monsoon conditions.
- Cut OpenWeather API calls by ~97% with Upstash Redis caching (30-min TTL for current weather, 3-hr for forecasts) — staying within free-tier limits across a full fleet.
- Implemented a 4-state alert machine (NORMAL → WARNING → CRITICAL → OVERDUE) with Firebase FCM push notifications.
- Built a zero-cost Route Optimizer using the Haversine formula ($\times 1.3$ urban factor) — no Google Maps API dependency.
- 50 automated tests across backend (18 Jest) and frontend (32 Vitest) with fully mocked external services in CI.

WORK EXPERIENCE

Cloud IT Support (Probation) · **CloudOps Solutions Inc.** | Waterloo, Canada Apr 2025 – May 2025

- Gained introductory exposure to GCP — observing cloud service deployment, basic IAM configuration, and environment monitoring.
- Assisted in troubleshooting minor cloud issues and navigating GCP console and documentation in a live operational context.

Software Developer (Fresher) · **Infosys Limited** | Waterloo, Canada

Sep 2023 – May 2024

- Learned C# and the .NET ecosystem on the job — progressed from OOP and ASP.NET Core Web API fundamentals to delivering a production Razor Pages application (EF Core 10, PostgreSQL, soft delete, pagination, Bootstrap 5) containerized with Docker and deployed to Render.
- Assigned by manager to independently design and deliver the EMT & B2B Pricing Management System — dual-module internal platform (Node.js, Express, PostgreSQL) for 6 B2B pricing types, deployed to Render and Railway.
- Applied JWT authentication, RBAC middleware, EF Core ORM, and GitHub Actions CI/CD across projects; worked within enterprise architecture patterns and team code review processes.

IT Intern · Royal Bank of Canada (RBC) | Waterloo, Canada

Jun 2023 – Sep 2023

- Performed manual and regression testing on internal banking apps, documenting defects and verifying fixes.
- Used Git/GitHub for version control — gaining practical experience with branching workflows in a team environment.
- Observed enterprise SDLC processes for the first time in a professional setting.

IT Intern / Data Analyst · Toronto-Dominion Bank (TD Bank Group) | Toronto, Canada

Jun 2020 – Sep 2020

- Supported data validation using SQL queries and Excel/CSV workflows for internal operations reporting.
- Assisted with repetitive data processing tasks using basic Python scripting.
- Gained first exposure to enterprise IT standards: documentation, data governance, and security awareness in a regulated environment.

EDUCATION

Bachelor of Science — Computer Science | Wilfrid Laurier University, Waterloo, Canada Sep 2021 – Present

Bachelor of Science — Computer Engineering Technology | Humber College, Toronto, Canada Sep 2018 – May 2021

CERTIFICATIONS

[Claude Code in Action](#) — Anthropic

Mar 2026

[GitHub Copilot Certification Skill Path](#) — Codecademy

Mar 2026

TECHNICAL SKILLS

Languages: JavaScript (ES2022+), Python, C# (.NET), SQL

Frontend: React 18, Redux Toolkit, Vite, Tailwind CSS, Bootstrap 5, Material-UI, Leaflet, Axios

Backend: Node.js, Express, ASP.NET Core (Razor Pages + Web API), RESTful API design, JWT, Stripe Webhooks, Zod

ORM & Databases: EF Core 10, Sequelize ORM, PostgreSQL (raw SQL + ORM), SQL Server, Supabase (RLS), Upstash Redis

AI & 3rd-Party APIs: OpenAI GPT-4, OpenWeather API, Firebase FCM, Stripe Checkout + Webhooks

Infrastructure & DevOps: Docker, Docker Compose, Nginx, GitHub Actions CI/CD, Render, Vercel, Railway, Neon.tech, GCP (basic)

Testing: Jest, Supertest, Vitest — unit & integration tests, mocked external APIs in CI

Architecture: Controller → Service → Repository, MVC (Razor Pages), RBAC, soft delete pattern, Redis TTL caching, AI-as-validation-gate